Question 1

a) Under normal operation the rectifier end controls the direct current while the inverter end operates with a constant extinction advance angle γ.

AC-side voltage reduction at rectifier end

- Rectifier end direct voltage and hence the direct current would reduce
- Rectifier controls the current by reducing the gate delay angle α
- Once the gate delay angle is stuck at its minimum limit, the inverter takes over current control
- Inverter controls the current by reducing its direct voltage by increasing the extinction advance angle γ
- Increase in γ lowers the inverter power factor and results in higher reactive power consumption at the inverter (i.e. opposite) end

[1 mark for each point – 5 marks]

AC-side voltage reduction at inverter end

- With constant extinction advance angle, the inverter end direct voltage would reduce and the direct current would increase
- Rectifier controls the direct current by increasing the gate delay angle α
- Increase in α lowers the rectifier power factor and results in higher reactive power consumption at the rectifier (i.e. opposite) end

[1 mark for each point – 3 marks]

b)

$$\text{ESCR}=\frac{S-Q_c}{P_{DC}}=\frac{S-mP_{DC}}{P_{DC}}\geq \text{ESCR}_{\min} \Rightarrow P_{DC}\leq \frac{S}{m+\text{ESCR}_{\min}}$$

[2 marks]

Maximum rated capacity at the connection point with system #1

$$P_{DC1}\leq \frac{S_1}{m+\text{ESCR}_{\min}}=7.69\ \text{GW}$$

[0.5 mark]

Maximum rated capacity at the connection point with system #2

$$P_{DC2}\leq \frac{S_2}{m+\text{ESCR}_{\min}}=8.13\ \text{GW}$$

[0.5 mark]

Model Answers

Maximum rated capacity of the DC links

$P_{DC-max} = \min(P_{DC1}, P_{DC2}) = \mathbf{7.69\ GW}$

[1 mark]

c)

The equivalent reactance at the connection point of the two isolated AC systems

$$x_1 = \frac{V^2}{S_1} = \frac{400^2}{35 \times 10^3} = 4.57\ \Omega, \quad x_2 = \frac{V^2}{S_2} = \frac{400^2}{37 \times 10^3} = 4.32\ \Omega$$

[1 mark]

We need to consider the worst case i.e. when the AC transmission line reactance is $x_l = 7\ \Omega$

[1 mark]

The equivalent reactance at the connection point of the AC systems with the AC line i.e., $x_l = 7\ \Omega$

$x_1' = x_1 \parallel (x_l + x_2) = 3.26\ \Omega$

$x_2' = x_2 \parallel (x_l + x_1) = 3.15\ \Omega$

[2 marks]

Modified short circuit levels

$$S_1' = \frac{V^2}{x_1'} = 49.13\ \text{GVA}, \quad S_2' = \frac{V^2}{x_2'} = 50.83\ \text{GVA}$$

[1 mark]

Maximum total rated HVDC capacity at the connection point with system #1

$$P_{DC1}' \leq \frac{S_1'}{m + \text{ESCR}_{\text{min}}} = 10.8\ \text{GW}$$

[0.5 mark]

Maximum total rated HVDC capacity at the connection point with system #2

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$$P'_{DC2} \leq \frac{S'_2}{m + \text{ESCR}_{\text{min}}} = 11.17 \text{ GW}$$

[0.5 mark]

Maximum rated capacity of the two HVDC links

$$P'_{DC-max} = \min(P'_{DC1}, P'_{DC2}) = 10.8 \text{ GW}$$

[1 mark]

Maximum rated capacity of the second HVDC links is

$$P'_{DC-max} - P_{DC-max} = \mathbf{3.11\ GW}$$

[1 mark]

d)

$$\text{ESCR} = \frac{S - Q_c}{P_{DC}} \Rightarrow Q_c \leq S - P_{DC} \times \text{ESCR}_{\text{min}}$$

[2 marks]

Without the AC line in operation

$$Q_{c2} \leq S_2 - P_{DC-max} \times \text{ESCR}_{\text{min}} = \mathbf{6.23\ GVAr}$$

$$\frac{Q_{c2}}{P_{DC-max}} \times 100 = \mathbf{81\%}$$

[1 mark]

With the AC line in operation

$$Q'_{c2} \leq S'_2 - P'_{DC-max} \times \text{ESCR}_{\text{min}} = \mathbf{7.64\ GVAr}$$

$$\frac{Q'_{c2}}{P'_{DC-max}} \times 100 = \mathbf{70.1\%}$$

[1 mark]

The percentage values are more as system #2 has higher short circuit level than system #1 and the the maximum rated capacity of the HVDC link is actually constrained by system #1

[1 mark]

Model Answers

Question 2

a)

No of 6-pulse bridge per pole $B = 2$

Rated current

$$I_d = \frac{P_d/2}{V_d} = 1.83 \text{ kA}$$

[1 mark]

Inverter end maintains the rated voltage $V_{di} = 600$ kV while operating at the minimum extinction angle by adjusting the effective turns ratio of the converter transformer

[1 mark]

$$V_{d0i} = \frac{V_{di} + BR_cI_d}{\cos(\gamma_{min})} = \frac{3\sqrt{2}}{\pi} BT_iE_{ac} \Rightarrow T_i = \frac{(V_{di} + BR_cI_d)\pi}{3\sqrt{2}BE_{ac}\cos(\gamma_{min})}$$

[1 mark]

With $E_{ac} = E_{ac-rated} = 400$ kV

$$T_i = \frac{(V_{di} + BR_cI_d)\pi}{3\sqrt{2}BE_{ac-rated}\cos(\gamma_{min})} = \mathbf{0.602}$$

[1 mark]

$$V_{d0i} = \frac{3\sqrt{2}}{\pi} BT_iE_{ac-rated} = 650.15 \text{ kV}$$

Power factor of inverter

$$\cos\phi_i = \frac{V_{di}}{V_{d0i}} = 0.92$$

[1 mark]

Reactive power consumption of inverter

$Q_i = 2 \times V_{di} \times I_d \times \cos\phi_i = \mathbf{918.14\ MVAr}$

[1 mark]

Direct voltage at rectifier end

$V_{dr} = V_{di} + R_LI_d = 609.17$ kV

[1 mark]

Model Answers

Rectifier end maintains the rated current while operating at the nominal firing angle by adjusting the effective turns ratio of the converter transformer

$$T_r = \frac{(V_{dr} + BR_c I_d)\pi}{3\sqrt{2}BE_{ac-rated}\cos(\alpha)} = \mathbf{0.604}$$

[1 mark]

$$V_{d0r} = \frac{3\sqrt{2}}{\pi} BT_r E_{ac} = 652.79 \text{ kV}$$

Power factor of inverter

$$\cos\phi_r = \frac{V_{dr}}{V_{d0r}} = 0.93$$

[1 mark]

Reactive power consumption of rectifier

$$Q_r = 2 \times V_{dr} \times I_d \times \cos\phi_r = \mathbf{860.28\ MVAr}$$

[1 mark]

b)

Here the inverter continues to maintain the rated voltage while the rectifier maintains the rated current while operating with the nominal gate delay angle. The rectifier end AC system voltage and direct voltage stays the same. So both reactive power consumption and turns ratio of the converter transformer on the rectifier end remains the same as under normal condition.

[2 marks]

Reduced AC voltage at the inverter end

$$E_{ac-i}{}' = 0.90 \times 400 \text{ kV} = 360 \text{ kV}$$

[1 mark]

$$T_i{}' = \frac{(V_{di} + BR_c I_d)\pi}{3\sqrt{2}BE_{ac-i}{}'\cos(\gamma_{min})} = \mathbf{0.67}$$

[1 mark]

The turns ratio basically compensates for the reduction in AC system voltage keeping V_{d0i} same as in normal condition. So the reactive power consumption of the inverter remains the same.

[1 mark]

Model Answers

c)

Reduced AC voltage at the rectifier end

$E_{ac-r}{}' = 0.85 \times 400\ \text{kV} = 340\ \text{kV}$

Revised gate delay angle needed to maintain the direct current

$$\cos(\alpha) = \frac{(V_{dr} + BR_c I_d)\pi}{3\sqrt{2}BT_r E_{ac-r}{}'} > 1$$

[1 mark]

There is no gate delay angle that allows the rectifier to maintain the direct current. So the inverter takes over current control with a modified current reference.

$$I_d{}' = I_d\left(1 - \frac{I_m}{100}\right) = 1.5\ \text{kA}$$

[1 mark]

At the rectifier end, the gate delay angle is stuck at the minimum value - so the effective turns ratio is adjusted to maintain the rated direct voltage.

[1 mark]

$$T_r{}' = \frac{(V_{d-rated} + BR_c I_d{}')\pi}{3\sqrt{2}BE_{ac-r}{}'\cos(\alpha_{min})} = \mathbf{0.671}$$

[1 mark]

$$V_{d0r}{}' = \frac{3\sqrt{2}}{\pi}BT_r{}'E_{ac-r}{}' = 615.88\ \text{kV}$$

Power factor of rectifier

$V_{dr}{}' = V_{d-rated} = 600\ \text{kV}$

$$\cos\phi_r{}' = \frac{V_{dr}{}'}{V_{d0r}{}'} = 0.97$$

[1 mark]

Reactive power consumption of rectifier

$Q_r{}' = 2 \times V_{dr}{}' \times I_d{}' \times \cos\phi_r{}' = \mathbf{417.75\ MVAr}$

[1 mark]

Model Answers

Direct voltage at the inverter end

$V_{di}{}' = V_{dr}{}' - R_L I_d{}' = 592.48\ \text{kV}$

[1 mark]

At the inverter end, the effective turns ratio is adjusted to produce this direct voltage with minimum extinction angle.

$$T_i{}' = \frac{(V_{di}{}' + BR_c I_d{}')\pi}{3\sqrt{2}BE_{ac-rated}\cos(\gamma_{min})} = \mathbf{0.59}$$

[1 mark]

$$V_{d0i}{}' = \frac{3\sqrt{2}}{\pi} BT_i{}' E_{ac-rated} = 638.78\ \text{kV}$$

Power factor of rectifier

$$\cos\phi_i{}' = \frac{V_{di}{}'}{V_{d0i}{}'} = 0.928$$

[1 mark]

Reactive power consumption of rectifier

$Q_i{}' = 2 \times V_{di}{}' \times I_d{}' \times \cos\phi_i{}' = \mathbf{717.86\ MVAr}$

[1 mark]

Model Answers

Question 3

a)

(i) If the current of the converter is zero, there are no voltage drops across the filter impedances or the equivalent series resistance of the circuit. Therefore, the V_o^{DC}=500V and V_o^{AC}=230 V with an angle of 180º (ie in the opposite direction to V_{AC}).

[2 mark]

The voltage of the submodule has to be enough to generate the summation of the DC voltage and the peak of the AC voltage, this gives $V_{SM} \geq V_{DC} + \sqrt{2}V_{AC} = 825\ V$

[1 mark]

(ii)

ii.a – The condition is that the power exchanged using DC current should be equal to the active power exchanged using AC current with opposite sign. In this case, we are told that the active power going to the AC grid is 3 kW, this means that the submodule must deliver 3 kW plus the losses incurred in the circuit using AC current. The losses can be neglected because the resistance of the circuit is about 0.02 pu (Zbase=13.2 Ω, R=0.023 pu) and the power setpoint is about 0.7 pu. The submodule achieves zero net power balance by drawing 3 kW using DC current.

[3 marks]

ii.b –We can calculate the AC current as:
$I_{AC} = \frac{P}{V_{AC}} = 13$ A flowing towards the AC grid.
Next, if we neglected the resistance of the circuit, we could simply calculate the DC current as:
$I_{DC} = \frac{P}{V_{DC}} = 6$A flowing towards the converter.
If we decided to carry out the calculation considering the resistance, we would find the AC voltage of the converter as:
$\bar{V}_o = -V_{AC} - (R + jX)\bar{I}_{AC} = 236.2$ V with angle -172º.
Then, the actual active power output of the submodule would be calculated as:
$P_o^{AC} = \Re e\{\bar{V}_o(\bar{I})^*\} = 3.05$ kW (out of the submodule).
If we don't neglect the resistance of the circuit, we must solve the following equations:
$V_{DC} - V_o^{DC} = R\ I_{DC}$ and $P_o^{DC} = V_o^{DC} I_{DC}$
This gives:
$I_{DC} = \frac{V_{DC} - \sqrt{V_{DC}^2 - 4P_o^{DC}R}}{2R} = 6.1$ A.
The error caused by neglecting the resistance is very small, therefore both procedures will be considered valid. However, full marks will only be awarded to those that neglect the resistance if they explain and justify that they neglect the resistance.

[4 marks]

ii.c – This question is closely related to the previous one, again it is possible to neglect the effect of the resistance of the circuit, which simplifies the solution of the DC voltage. Following the previous question we can simply do:
$V_o^{DC} = V_{DC} = 500$ V.

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The calculation of VoDC requires calculating the voltage drop across X, which would cause a much bigger error if neglected (still possible if argued properly):
$\bar{V}_o = -V_{AC} - (R + jX)\bar{I}_{AC} = 236.2$ V with angle -172º.

If R wasn't neglected in the calculation of V_o^{DC}, we would use the equations we derived in the previous question:

$V_o^{DC} = \frac{V_{DC}+\sqrt{V_{DC}^2-4P_{oDC}R}}{2} = 498$ V.

Again, answers that neglected the resistance will be considered to be correct as long as the candidates justify the simplification.

[4 marks]

(iii)

iii.a – The transfer function for Gd1 is:

$$G_{d1} = -\frac{1}{Ls + R}$$

And for Gd2, it is

$$G_{d2} = \frac{1}{Ls + R}$$

These come from the equation of the circuit, which is $V_{DC} - V_o - V_{AC} = R\, i + L\, \frac{d}{dt} i$

Marks will be subtracted if the equation of the circuit isn't given and the signs of the transfer functions aren't consistent with the definition of the variables in the circuit diagram.

[3 marks]

iii.b – The current in this circuit contains DC plus AC of 50 Hz and the external disturbance affecting this current are the AC grid voltage and the DC grid voltage, which are AC of 50 Hz and DC respectively. The transfer function between the current and the control action, V_o, is a first order. The disturbances can be measured and a feed-forward can be used to cancel out their effect. A PI controller will provide zero steady-state error for tracking DC current references but it would give steady-state error for the AC component of the current reference. This could be mitigated by replacing the PI or by using a reference pre-filter if the phase-delay and the decay of gain at 50 Hz could be estimated accurately.

[2 marks]

b)

(i)

The base impedance can be calculated as: $Z_{base} = \frac{V_{AC}^2}{P} = 7.2\ \Omega$. Then, the pu impedances are calculated as:

$$R_{pu} = \frac{R}{Z_{base}}$$

And

$$X_{pu} = \frac{X}{Z_{base}}$$

For the converter parameters given here, we get R_{arm}=R_L= 0.01 pu, X_{arm}= 0.2 pu and X_L= 0.05 pu.

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[2 marks]

The series resistance looks fine because it implies losses fo about 1%, which is consistent with the requirement of high efficiency. Larger values will cause large losses in the converter, which wouldn't be acceptable at transmission scale.
The inductance of the arms is well aligned with the common values used in practice. The equivalent inductance seen in the equivalent differential circuit is of 0.15 pu, the inductance in the equivalent summative circuit is of 0.4 pu. A large inductance in the equivalent differential circuit would draw large reactive power and more importantly it could limit the area of operation due to the limitation of the voltage generation capability of the converter. A low inductance value would cause the dynamics to be much faster, which would be problematic when dealing with short-circuits, they would make the reaction time to interrupt the current be shorter. The question asked explicitly about increasing the impedance, therefore no comments are expected from the candidates about the consequences of reducing it.

[2 marks]

(ii)

The summative current is defined as $I_{sum} \triangleq \frac{1}{2}I_u + \frac{1}{2}I_l$, therefore $\bar{I}_{sum} = 7.5\ A$ and $I_{sum}{}^{DC}$= 25 A. The differential current, or current going into the AC grid is $I_{AC} = I_u - I_l$, which gives $\bar{I}_{AC} = 35$ A and $I_{AC}{}^{DC}$= -20 A. Normally the DC component of the AC grid current should be zero but it is physically possible to make it different than zero.

[1.5 marks]

The summative DC current is used to exchange power between the DC grid and the converter, which is required in order to compensate for the power that the converter exchanges with the AC grid using differential AC current. The AC component of the summative current is only used for control purposed following transients; it causes an exchange of active power between the upper stack and the lower stack and this can be used to rebalance the energy of the two.

[1.5 marks]

(iii)

iii.a – The question doesn't tell us the power that is exchanged using DC current but it does tell us that there's no DC current going into the AC grid. Therefore we can infer that the power the upper stack and the lower stack exchange using DC current is the same and equal to 52.5 MW (there are no losses in the circuit).

[2 marks]

iii.b – The question tells us that the converter draws 105 MW from the DC grid but it delivers 100 MW to the AC grid, this causes a surplus of 5 MW internally and we are told that the losses are neglected. Consequently, the converter is building up internal energy, which implies that this isn't a stable operating point. The question also tells us that the upper stack delivers 60 MW using AC current

Model Answers

whereas the lower stack only delivers 40 MW. This implies that the net power balance of the upper stack is of -7.5 MW whereas the lower stack has a surplus of 12.5 MW. This means that the upper stack is discharging over time whereas the upper stack is building up energy over time.

[2 marks]

Model Answers